

Disk Management

Dasar Teknologi Informasi

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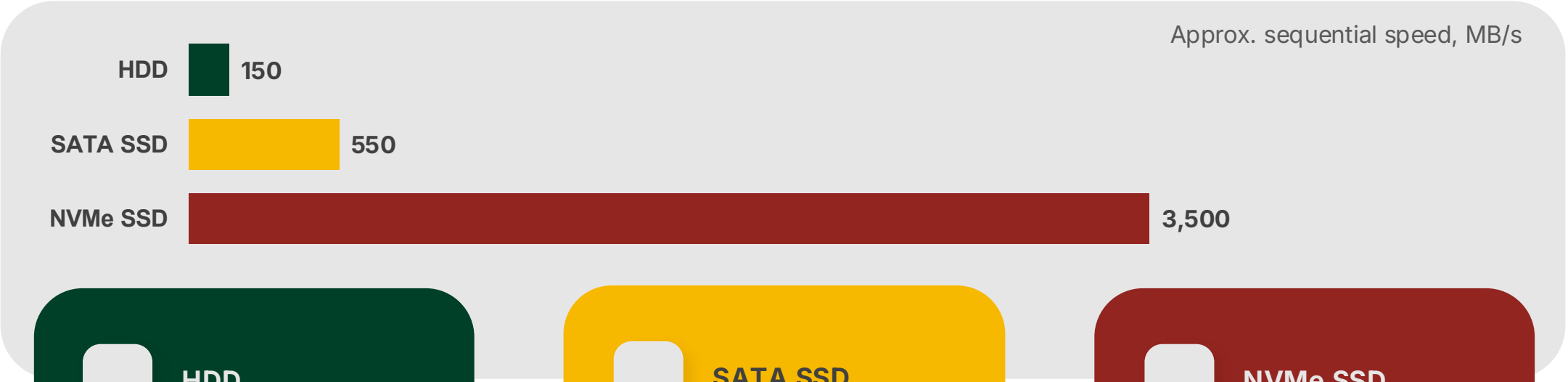
Learning Objectives

By the end of this session, you will be able to:

1. Distinguish storage media: HDD, SSD, NVMe, and flash
2. Explain partitions, MBR vs GPT, and file systems
3. Manage disks with Disk Management and diskpart
4. Describe RAID basics, disk maintenance, and common troubleshooting

Today's Agenda

1. Storage media and interfaces: HDD, SSD, NVMe, SATA, M.2
2. Partitions, MBR vs GPT, and file systems
3. Disk Management and diskpart
4. Volume operations and hands-on practice
5. RAID basics
6. Maintenance and troubleshooting



HDD

Spinning magnetic platters. Large and cheap, but slower.

SATA SSD

Flash memory over SATA. A big upgrade for older PCs.

NVMe SSD

Flash memory over PCIe. Best for the OS and heavy apps.

Storage Media

Speeds are rough typical values; real products vary.

HDD vs SSD

HDD

- Parts: platters, spindle motor, read/write heads, actuator arm
- Moving parts make it noisier and sensitive to shocks
- Low price per GB and very large capacities
- Slower access: heads must move to the data

SSD

- Parts: a controller and NAND flash memory chips
- No moving parts: silent and shock-resistant
- Much faster access and a more responsive system
- Higher price per GB and limited write endurance (TBW)

Storage Interfaces

Interface	Form factor	Notes
SATA III	Separate data and power cables; 2.5-inch or 3.5-inch drives	Theoretical limit 6 Gb/s (about 550 MB/s)
M.2	Small card plugged directly into a motherboard slot	Form factor only: can carry SATA or NVMe
NVMe	Protocol over PCIe lanes, usually on an M.2 card	Much faster than SATA
USB	External drives and flash drives	Speed depends on the USB version (2.0, 3.x)

M.2 is a form factor. NVMe is a protocol.

MBR disks

Primary

Can hold the OS and be used to boot the computer.

Max 4 per disk

MBR disks

Extended

A container that holds logical partitions. Only one per disk.

1 per disk

MBR disks

Logical

Created inside the extended partition to add more drives.

Inside extended

Disk Partitions

Partitions split one physical disk into logical areas: separate the OS from data, enable multi-boot, simplify backups.

GPT is simpler

No primary, extended, or logical distinction. Windows supports up to 128 partitions.

MBR vs GPT

Aspect	MBR	GPT
Disk size	Up to 2 TB	Over 2 TB (practically unlimited)
Partitions	4 primary, or 3 primary + 1 extended	Up to 128 in Windows
Boot mode	Legacy BIOS	UEFI
Compatibility	Very wide, including old systems	Modern systems; Windows 11 needs UEFI

Converting MBR to GPT in Disk Management is only available on an empty disk. For disks with data, use mbr2gpt.

File Systems

File system	Platform	Key features
FAT32	Almost all devices	Wide compatibility; maximum file size 4 GB
exFAT	Windows, macOS, Linux	Good for large flash drives and SD cards; no journaling
NTFS	Windows (system drive)	Permissions, encryption, compression, journaling
ext4	Linux	Journaling; default on many Linux distributions
APFS	macOS and iOS	Snapshots, encryption, tuned for SSDs

Quick Format only builds new structures; the old data is not truly erased and can often be recovered.

Two Tools for Managing Disks

Disk Management

- Open: right-click Start, then choose Disk Management
- Or press Win + R and type diskmgmt.msc
- Common statuses: Online, Unallocated, Not Initialized
- Every action is a right-click on a disk or volume

diskpart

Run Command Prompt as Administrator

```
diskpart
list disk
select disk 1
clean
convert gpt
create partition primary
format fs=ntfs quick
assign letter=E
```

Warning: clean erases the entire disk. Double-check the disk number first.

Example disk layout in Disk Management



Preparing a New Disk

A brand-new disk shows up in Disk Management as Not Initialized or Unallocated.

STEP 01

Initialize Disk

Right-click the new disk (Not Initialized) and choose GPT for modern PCs or MBR for old ones.

STEP 02

New Simple Volume

Right-click the Unallocated space, set the size, then assign a drive letter or mount folder.

STEP 03

Format Volume

Pick NTFS or exFAT, add a label, and tick Quick Format. Then click Finish.

Common Volume Operations

Erases existing data

Format

Creates a new file system on the volume.

Needs adjacent space

Extend Volume

Adds space; needs Unallocated space right next to the volume.

Frees up space

Shrink Volume

Makes a volume smaller to create Unallocated space.

Drive letter or folder

Change Letter

Changes the drive letter or mounts it in an empty NTFS folder.

Erases the volume

Delete Volume

Removes the volume; its space becomes Unallocated.

MBR to GPT

Convert Disk

Greyed out unless the disk is empty. Use mbr2gpt for disks with data.

RAID Basics

Level	Min. disks	How it works	Fault tolerance
RAID 0 (striping)	2	Splits data across all disks for speed	None: one failed disk loses everything
RAID 1 (mirroring)	2	Writes identical copies to two disks	Survives 1 failed disk
RAID 5	3	Striping with distributed parity	Survives 1 failed disk
RAID 10	4	Mirrors first, then stripes across mirrors	Survives 1 failed disk per mirror pair

RAID is not a backup. It does not protect against deletion, malware, or fire.

Disk Maintenance

Regular backups

Follow the 3-2-1 rule: 3 copies, 2 media types, 1 copy off-site.

Check disk health

Monitor SMART status and replace the disk when warnings appear.

Check for file errors

Run chkdsk to find and repair file system errors.

Optimize by disk type

HDD: defragment regularly. SSD: rely on TRIM (Optimize Drives); no defragmentation needed.

Troubleshooting Common Problems

Symptom	Likely cause	What to do
New disk does not appear	Not initialized or formatted	Initialize, create a volume, assign a drive letter
Disk not detected at all	Loose cable, bad port, or disabled in BIOS/UEFI	Check cables and ports, then check BIOS/UEFI
HDD makes clicking noises	Mechanical parts failing	Back up immediately, then replace the disk
Disk at 100% and system slow	Old HDD, full disk, or too many processes	Check SMART, clean up, consider an SSD
Disk shows as RAW	Damaged or unrecognized file system	Recover data first, then format again

Summary

1. HDDs are cheap and large, SSDs are fast and silent. M.2 is a shape, NVMe is a protocol.
2. A disk is divided into partitions: GPT for modern disks, MBR for legacy systems.
3. A partition needs a file system (NTFS, exFAT, ...) before it can store files.
4. Disk Management and diskpart manage volumes: be careful with Format and clean.
5. RAID is not a backup: back up regularly and check disk health.

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